

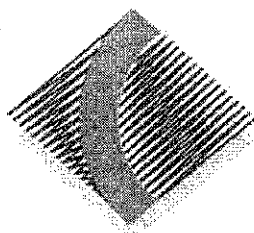
JG
HK
KL
AT

Name: _____

Class: 12MTX_____

Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2012 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 1

*Time allowed - 2 HOURS
(Plus 5 minutes reading time)*

Directions to candidates

- Attempt all questions
- Approved calculators may be used.
- Standard Integral Tables are provided at the back of this paper.
- Write your name and class in the space provided at the top of this question paper.

Section I - TOTAL MARKS 10

- To be answered on the removable answer grid at the back of the exam paper.
- Allow about 15 minutes for this section.

Section II - TOTAL MARKS 60

- All answers to be completed on the writing paper provided. Each question is to be commenced on a new page clearly marked Question 11, Question 12, etc on the top of the page. Write your name and class at the top of each page.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Allow about 1 hour and 45 minutes for this section.

YOUR ANSWERS WILL BE COLLECTED IN ONE BUNDLE. THE MULTIPLE CHOICE SECTION I ON TOP AND THEN WRITTEN ANSWERS TO SECTION II AND THEN THE QUESTION PAPER.

SECTION I 10 MARKS**INSTRUCTIONS**

- Attempt all questions
 - Allow about 15 minutes for this section
 - Section I answers are to be completed on the multiple-choice answer sheet attached to the back of this question paper.
 - Select the alternative A, B, C or D that best answers the question
-

1. What is the acute angle between the lines $y = 2x - 1$ and $x - 3y + 6 = 0$?
(A) 18°
(B) 45°
(C) 63°
(D) 82°

2. Let α , β and γ be the roots of $x^3 - x^2 + cx + 12 = 0$. It is known that two roots are equal in magnitude but opposite in sign. What is the value of c ?
(A) -12
(B) $-2\sqrt{3}$
(C) $2\sqrt{3}$
(D) 12

3. $P(x) = x^3 + 4x^2 - 5x + 4$ divided by $x - 2$, expressed in the form of $P(x) = Q(x).A(x) + R(x)$ is
(A) $P(x) = (x - 2)(x^2 + 2x - 1) + 2$
(B) $P(x) = (x - 2)(x^2 + 6x - 17) - 30$
(C) $P(x) = (x - 2)(x^2 + 6x + 7) + 18$
(D) $P(x) = (x - 2)(x^2 + 2x - 9) - 14$

4. The curve $y = \sin x$ is rotated about the x -axis from $x = 0$ to $x = \frac{\pi}{2}$. Find the volume of the solid formed.
(A) $\frac{\pi}{4}(\pi - 1)$ units³
(B) $\frac{\pi^2}{4}$ units³
(C) $\frac{\pi}{2}$ units³
(D) $\frac{\pi}{4}$ units³

5. If the velocity v of a particle moving on the x -axis is given by

$$v^2 = -3x^2 + 20x + 7$$

Which of the following expresses its acceleration in terms of x ?

(A) $\ddot{x} = -3\left(x - 3\frac{1}{3}\right)$

(B) $\ddot{x} = -3(x - 2)$

(C) $\ddot{x} = -3(x - 3)$

(D) $\ddot{x} = -2(x - 2)$

6. Find the sum of the coefficients of $(1 + x)^{16}$

(A) 131 072

(B) 65 536

(C) 17

(D) 32 768

7. Evaluate $\int_e^{e^2} \frac{dx}{x \ln 2}$

(A) $\ln 2$

(B) $\frac{1}{\ln 2}$

(C) $\frac{-1}{2e^2}$

(D) $\frac{1}{2e^2}$

8. It is known that $\int_0^4 f(x) dx = 6$. Hence, the value of $\int_3^7 f[(x-3)+2] dx$ is

(A) 8

(B) 18

(C) 14

(D) 16

9. If $\frac{dN}{dt} = 0.1(N - 100)$ and $N = 300$ when $t = 0$, which of the following is true?

(A) $N = 200 + 100e^{0.1t}$

(B) $N = 300 + 100e^{0.1t}$

(C) $N = 100 + 200e^{0.1t}$

(D) $N = 100 + 300e^{0.1t}$

10. If $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$,

then the expression $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n} =$

(A) 0

(B) $n(2)^{n-1}$

(C) 1

(D) 2^n

END OF SECTION 1

SECTION II 60 MARKS

INSTRUCTIONS

- Answer all questions on the writing paper provided
- Allow about 1 hour and 45 minutes for this section
- Begin each question on a new page.
- Show all necessary working.

Question 11 (15 marks)

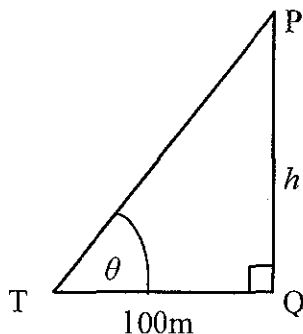
BEGIN A NEW PAGE

Marks

- (a) A particle moves in a straight line and its position x metres at time t seconds is given by

$$x = 2 + \sin 4t + \sqrt{3} \cos 4t.$$

- (i) By first expressing $\sin 4t + \sqrt{3} \cos 4t$ in the form of $R \sin(4t + \alpha)$, prove that it is undergoing simple harmonic motion. 4
- (ii) Find the equilibrium position and the amplitude of the motion. 1
- (iii) Find the maximum speed of the particle. 1
- (b) A parachutist, P, jumps out of a plane at height $h(t)$ metres above the ground and by the time he reaches 3000 m, he is falling at a constant rate of 5.5 m/s. Point Q is on horizontal ground directly below him. An observer at T is 100 m from Q and the angle of elevation from this point to the parachutist is $\theta(t)$ radians.



- (i) Show that $\frac{dh}{d\theta} = \frac{100}{\cos^2 \theta}$. 1
- (ii) Show that the rate of decrease of the angle of elevation when $h = 2000$ m is 0.000137 rad/s. 3
- (c) Use the substitution $u = \tan \theta$ to find the exact value of this integral 2

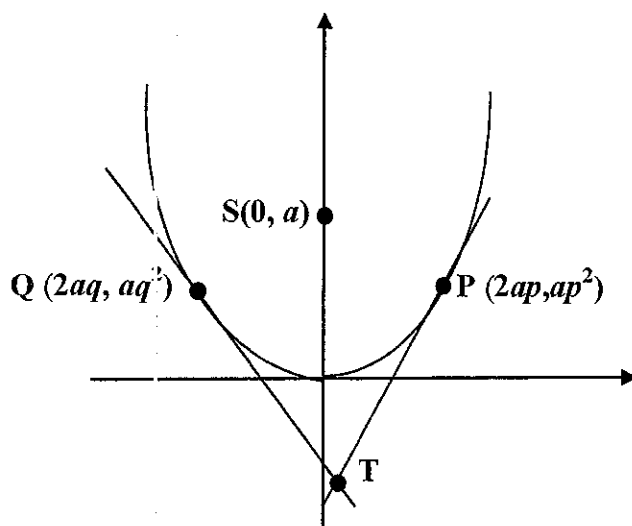
$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 \theta}{\tan \theta} d\theta.$$

- (d) Find all values of x for which $\frac{6}{x} \geq x - 1$. 3

Question 12 (15 marks) BEGIN A NEW PAGE

Marks

- (a) Differentiate $[1 + \cos^{-1}(3x)]^3$ with respect to x . 2
- (b) Write down the general solution of $\sqrt{3} \tan \theta - 1 = 0$.
Leave your answer in exact radian form. 1
- (c) (i) Rewrite $-3 - x^2 - 4x$ in the form $b^2 - (x + a)^2$ where a and b are integers. 1
- (ii) Hence, or otherwise, evaluate $\int \frac{dx}{\sqrt{-3 - x^2 - 4x}}$. 1
- (d) Consider the function $f(x) = \sin^{-1}(x - 1)$.
- (i) State the domain and range of $y = f(x)$. 2
- (ii) Draw the graph of $y = f(x)$. 1
- (iii) The area bounded by the curve $y = f(x)$, the y -axis and the line $y = \frac{\pi}{2}$ is rotated about the y -axis. Find the volume of the solid formed. 2
- (e) Consider the parabola $4ay = x^2$ where the focal length is a , ($a > 0$), and the tangents at $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ intersect at the point T .
Let $S(0, a)$ be the focus of the parabola.
- (i) Find the coordinates of T .
(You may assume that the equation of the tangent at P is $y = px - ap^2$) 1
- (ii) Show that the length $SP = a(p^2 + 1)$ 1
- (iii) Suppose P and Q move on the parabola in such a way that $SP + SQ = 4a$. Show that T is constrained to move on a parabola. 3



- (a) A particle is projected from the top of a cliff 200m high. The horizontal and vertical components of the velocity when $t = 0$ are $20\sqrt{3} \text{ m/s}$ and 30 m/s respectively.

(i) Determine the parametric equations of the path of the stone after t seconds.
(take $g = 10 \text{ m/s}^2$)

2

(ii) Find when the particle hits the ground.

2

(iii) Find the velocity and the angle of impact of the particle when it hits the ground.

3

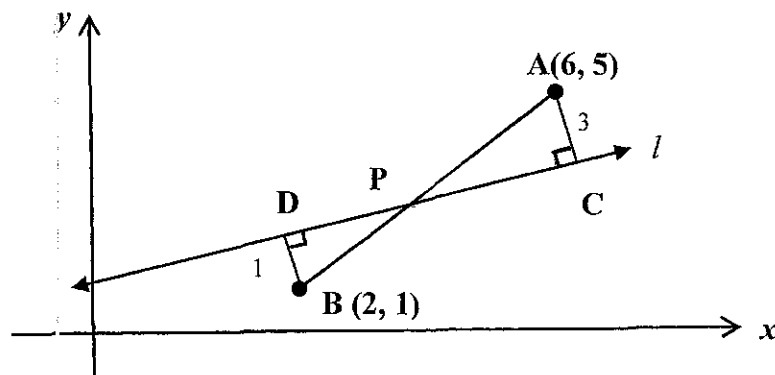
- (b) Evaluate $\lim_{t \rightarrow 0} \left[\frac{4 \sin^2 t}{t^2} \right]$

1

- (c) The equation $2 \cos^3 \theta - \cos^2 \theta + \cos \theta - 1 = 0$ has solutions $\cos a$, $\cos b$ and $\cos c$. Prove that $\sec a + \sec b + \sec c = 1$.

2

- (d)



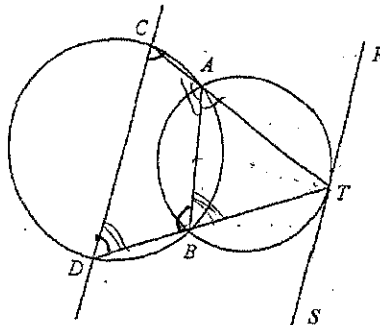
The points $A(6, 5)$ and $B(2, 1)$ are 3 units and 1 unit respectively from the line l and are on opposite sides of l , as shown in the diagram.

Find the coordinates of the point P , where the interval AB crosses the line l .

3

- (e) Two unequal circles intersect at A and B . The line RS is a tangent to the smaller circle at T . The lines TA and TB meet the larger circle at C and D respectively. Prove that $RS \parallel CD$.

2



(a) (i) Prove that $\tan(x + h) - \tan x = \frac{\sin h}{\cos(x + h) \cos x}$. 2

(ii) Hence, find the derivative of $\tan x$ from first principles. 1

(b) (i) Show that $(1 + x)^{2n} \left(1 - \frac{1}{x}\right)^{2n} = \left(x - \frac{1}{x}\right)^{2n}$. 1

(ii) Hence, by equating the constant terms, deduce that

$$({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + ({}^{2n}C_{2n})^2 = (-1)^n {}^{2n}C_n.$$
2

(c) (i) Show that the equation $2x^3 - 3x^2 + 0.999 = 0$ has a root near $x = 1$. 1

(ii) Explain why Newton's method fails if the first approximation taken for $2x^3 - 3x^2 + 0.999 = 0$ is $x = 1$. 2

(iii) Using $x = 1.5$ find, by one application of Newton's method, a better approximation of the root of the equation $2x^3 - 3x^2 + 0.999 = 0$. 2

(d) Prove by mathematical induction, $\sum_{j=1}^n \sin(2j - 1)x = \frac{1 - \cos 2nx}{2 \sin x}$ 4

Hint : You may find $\sin(2k + 1)x = \sin(2kx + x)$ useful.

END OF THE PAPER

Ext 1 AP4
MCQ ANSWERS

- 1) B
- 2) A
- 3) C
- 4) B
- 5) A
- 6) B.
- 7) B.
- 8) C
- 9) C
- 10) B.

Q11)

a) (i) $x = 2 + \sin 4t + \sqrt{3} \cos 4t$

$$\begin{aligned}\sin 4t + \sqrt{3} \cos 4t &= R \sin(4t + \alpha) \\ &= R [\sin 4t \cos \alpha + \cos 4t \sin \alpha] \\ &= R \cos \alpha \sin 4t + R \sin \alpha \cos 4t\end{aligned}$$

$$\therefore \left. \begin{array}{l} R \cos \alpha = 1 \\ R \sin \alpha = \sqrt{3} \end{array} \right\} \tan \alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}$$

(Both R and α) ✓

$$\begin{aligned}R^2 \sin^2 \alpha + R^2 \cos^2 \alpha &= 3 + 1 = 4 \\ R^2 &= 4 \Rightarrow R = 2\end{aligned}$$

$$\begin{aligned}x &= 2 + \sin 4t + \sqrt{3} \cos 4t \\ x &= 2 + 2 \sin(4t + \frac{\pi}{3}) \\ \dot{x} &= 4 \times 2 \cos(4t + \frac{\pi}{3}) \\ \ddot{x} &= -16 \times 2 \sin(4t + \frac{\pi}{3})\end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \checkmark$$

$$\ddot{x} = -16(x-2) \quad \checkmark \text{ as } x-2 = 2 \sin(4t + \frac{\pi}{3})$$

$\therefore$ It is undergoing SHM.

(ii) $\left. \begin{array}{l} \text{Equilibrium position} = 2 \\ \text{amplitude} = 2 \end{array} \right\} \checkmark$

(iii) $\text{max. speed} = 8 \text{ ms}^{-1} \quad \checkmark$

Q11) (b)

$$(i) \tan \theta = \frac{h}{100}$$

$$\therefore h = 100 \tan \theta$$

$$\frac{dh}{d\theta} = 100 \sec^2 \theta = \frac{100}{\cos^2 \theta}$$

$$(ii) \frac{d\theta}{dt} = \frac{d\theta}{dh} \times \frac{dh}{dt} \quad \text{given } \frac{dh}{dt} = -5.5$$

$$\text{When } h = 2000, \tan \theta = \frac{h}{100}$$

$$\therefore \theta = \tan^{-1}(20) = 1.52084$$

$$\therefore \frac{d\theta}{dt} = \frac{\cos^2(1.52084)}{100} \times -5.5 \quad \checkmark$$

$$= -0.000137 \text{ rad s}^{-1} \quad \checkmark \quad (-1 \text{ for not showing negative,})$$

$$\therefore \text{rate of decrease} = 0.000137$$

$$(c) \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 \theta}{\tan \theta} d\theta$$

$$u = \tan \theta$$

$$du = \sec^2 \theta d\theta$$

$$\theta = \frac{\pi}{3}, u = \sqrt{3}$$

$$\theta = \frac{\pi}{6}, u = \frac{1}{\sqrt{3}}$$

$$= \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{du}{u} \quad \checkmark$$

$$= \left[\ln u \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}$$

$$= \ln \sqrt{3} - \ln \frac{1}{\sqrt{3}}$$

$$= \ln \left(\frac{\sqrt{3}}{\frac{1}{\sqrt{3}}} \right) = \ln 3 \quad \checkmark$$

Q11) (d)

$$\frac{6}{x} \geq x-1 \quad (x \neq 0)$$

$$(xx^2) \quad 6x \geq x^2(x-1) \quad \checkmark$$

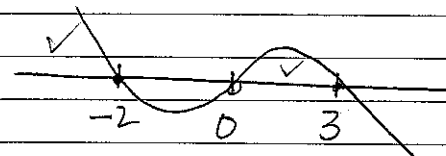
$$6x - x^2(x-1) \geq 0$$

$$x[6 - x(x-1)] \geq 0$$

$$x[6 - x^2 + x] \geq 0$$

$$x(6 + x - x^2) \geq 0$$

$$x(2+x)(3-x) \geq 0 \quad \checkmark$$



$$x \leq -2, 0 \leq x \leq 3 \quad \checkmark$$

$$(-1 \text{ for } 0 \leq x \leq 3)$$

Q12)(a)

$$\frac{d}{dx} [1 + \cos^{-1}(3x)]^3 = 3[1 + \cos^{-1}(3x)]^2 \left[\frac{-3}{\sqrt{1-9x^2}} \right]$$

(b) $\tan \theta = \frac{1}{\sqrt{3}}$

$\therefore \theta = n\pi + \frac{\pi}{6} \quad \checkmark$

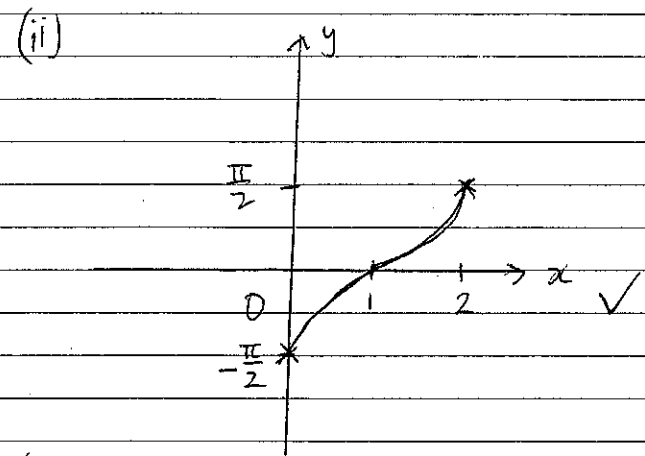
(c) (i) $-\left[x^2 + 4x + 3\right] = -\left[x^2 + 4x + 2^2 - 2^2 + 3\right]$
 $= -\left[(x+2)^2 - 1\right]$
 $= 1 - (x+2)^2 \quad \checkmark$

(ii) $\int \frac{dx}{\sqrt{-3-x^2-4x}} = \int \frac{dx}{\sqrt{1-(x+2)^2}}$
 $= \sin^{-1}(x+2) + C \quad \checkmark$

Q12) d) $f(x) = \sin^{-1}(x-1)$

(i) Domain $-1 \leq x-1 \leq 1$
 $0 \leq x \leq 2 \quad \checkmark$

Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \quad \checkmark$



(iii)

$$\begin{aligned} \text{Volume} &= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 dy & y &= \sin^{-1}(x) \\ &= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin y + 1)^2 dy & \sin y &= x-1 \\ & & \sin y + 1 &= x \\ &= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 y + 2\sin y + 1 dy \\ &= \pi \left[\frac{y}{2} - \frac{\sin 2y}{4} + y - 2\cos y \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \quad \checkmark \\ &= \pi \left[\frac{3y}{2} - \frac{\sin 2y}{4} - 2\cos y \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \quad \checkmark \end{aligned}$$

Q12) (e)

(i) $y = px - ap^2$ (tangent at P) — ①

$y = qx - aq^2$ (tangent at Q).

$\therefore px - ap^2 = qx - aq^2$

$(p - q)x = a(p^2 - q^2)$

$(p - q)x = a(p + q)(p - q)$

$x = a(p + q)$

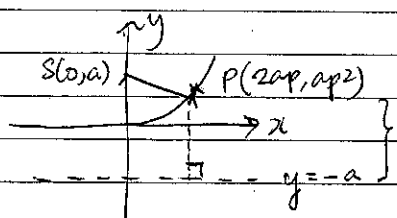
Sub. into ①

$y = pa(p + q) - ap^2$

$y = ap^2 + apq - ap^2 = apq.$

$\therefore T$ has coordinates $T(a(p + q), apq)$ ✓

(ii) Using definition,



$ap^2 - (-a) = ap^2 + a = a(p^2 + 1)$ ✓

(or use distance formula to find SP.)

12 (e) (iii)

$SP = a(p^2 + 1)$

Similarly, $SQ = a(q^2 + 1)$

$SP + SQ = 4a \Rightarrow a(p^2 + 1) + a(q^2 + 1) = 4a$
 $p^2 + q^2 = 2$ ✓

For $T(a(p + q), apq)$

$x = a(p + q)$ $y = apq$

$p + q = \frac{x}{a}$, $pq = \frac{y}{a}$

$(p + q)^2 = \frac{x^2}{a^2}$

$p^2 + 2pq + q^2 = \frac{x^2}{a^2}$

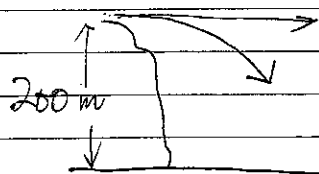
But $p^2 + q^2 = 2$, $2 + 2pq = \frac{x^2}{a^2}$

and $pq = \frac{y}{a}$, $2 + \left(\frac{2y}{a}\right) = \frac{x^2}{a^2}$

$2a^2 + 2ay = x^2$ ✓ which is a parabola.

Q13)

(a) (i)



$$y'' = -10$$

$$y' = -10t + C$$

$$t=0, y' = 30 = -10(0) + C$$

$$\therefore y' = 30 - 10t$$

$$y = 30t - 5t^2 + C_1$$

$$t=0, y = 200 = 0 + C_1$$

$$\therefore y = 30t - 5t^2 + 200 \quad \checkmark$$

$$x'' = 0$$

$$x' = C_2$$

$$t=0, x' = 20\sqrt{3} = C_2$$

$$\therefore x' = 20\sqrt{3}$$

$$x = 20\sqrt{3}t + C_3$$

$$t=0, x = 0 + C_3 = 0$$

$$\therefore x = 20\sqrt{3}t \quad \checkmark$$

(to gain 1 in all workings must be shown correctly.)

(ii) $y = 0$

$$30t - 5t^2 + 200 = 0$$

$$6t - t^2 + 40 = 0$$

$$t^2 - 6t - 40 = 0 \quad \checkmark$$

$$(t - 10)(t + 4) = 0$$

$$t = 10s \quad \checkmark$$

Q13 (a) (iii)

When $t = 10$,

$$x' = 20\sqrt{3},$$

$$y' = 30 - 10(10) = -70.$$

$$v^2 = x'^2 + y'^2 = 70^2 + 1200$$

$$\therefore v = -10\sqrt{61} \quad \checkmark \quad (\text{0 for not showing negative sign.})$$

$$\tan \theta = \frac{y'}{x'} = \frac{-70}{20\sqrt{3}} \quad \checkmark$$

$$\therefore \theta = \tan^{-1}(-2.0207...) = 116.3^\circ$$

$$= 116^\circ \quad \checkmark$$

(must be obtuse.)

(-1 for showing acute.)

Q13)b)

$$\lim_{t \rightarrow 0} \left[\frac{4 \sin t}{t} \cdot \frac{\sin t}{t} \right] = 4 \quad \checkmark$$

$$(c) \quad 2 \cos^3 \theta - \cos^2 \theta + \cos \theta - 1 = 0$$

$\cos a, \cos b, \cos c$ are roots.

$$\therefore \sec a + \sec b + \sec c$$

$$= \frac{1}{\cos a} + \frac{1}{\cos b} + \frac{1}{\cos c}$$

$$= \frac{\cos b \cos c + \cos a \cos c + \cos a \cos b}{\cos a \cos b \cos c} \quad \checkmark$$

$$= \frac{\text{Sum of roots 2 at a time}}{\text{Product of roots}}$$

$$= \frac{C/A}{-D/A} \quad \checkmark$$

$$= \frac{C}{-D} = \frac{1}{-(-1)} = 1$$

$$A = 2$$

$$B = -1$$

$$C = 1$$

$$D = -1$$

1 - showing
sum of roots
two at a
time

1 - showing
product of
roots

Q13)

$$\begin{aligned} d) \quad \angle BDP &= \angle ACP = 90^\circ \text{ (given)} \\ \angle DPB &= \angle CPA \text{ (Vertically opp. \(\angle\)'s)} \\ \therefore \triangle DPB &\parallel \triangle CPA \text{ (equiangular)} \quad \checkmark \end{aligned}$$

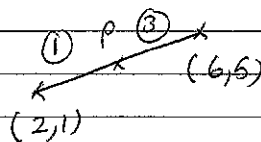
$$\frac{DB}{CA} = \frac{DP}{CP}$$

$$\frac{1}{3} = \frac{DP}{CP} \quad \therefore DP:CP = 1:3 \quad \checkmark$$

(show
all work
for
similarity)

$$P(x, y) = \left(\frac{2(3) + 6(1)}{4}, \frac{1(3) + 5(1)}{4} \right)$$

$$= (3, 2) \quad \checkmark$$



$$\begin{aligned} e) \quad \angle RTA &= \angle ABT \text{ (alt seg. theorem)} \\ \angle ABT &= \angle ACD \text{ (ext. \(\angle\) of cyclic quad.)} \end{aligned} \quad \checkmark$$

$$\begin{aligned} \text{As } \angle ACD &= \angle RTA, \\ CD &\parallel RS \text{ (equal alternate angles)} \quad \checkmark \\ &\text{imply parallel lines} \end{aligned}$$

Q14 a) (i) $\tan(x+h) - \tan x = \frac{\sinh}{\cos(x+h)\cos x}$

$$\begin{aligned}
 \text{LHS} &= \frac{\tan x + \tanh}{1 - \tan x \tanh} - \tan x \\
 &= \frac{\cancel{\tan x} + \tanh - \cancel{\tan x} + \tan^2 x \tanh}{1 - \tan x \tanh} \\
 &= \frac{\tanh + \tan^2 x \tanh}{1 - \tan x \tanh} \quad \checkmark \\
 &= \frac{\frac{\sinh}{\cosh} (1 + \tan^2 x)}{1 - \tan x \left(\frac{\sinh}{\cosh} \right)} \\
 &= \frac{\sinh (1 + \tan^2 x)}{\cosh - \tan x \sinh} \\
 &= \frac{\sinh \left(1 + \frac{\sinh^2 x}{\cos^2 x} \right)}{\cosh - \frac{\sinh x}{\cos x} \sinh} \\
 &= \frac{\sinh (\cos^2 x + \sinh^2 x)}{\cosh \cos^2 x - \sinh x \sinh \cos x} \\
 &= \frac{\sinh}{\cos x (\cosh \cos x - \sinh^2 \sinh)} \quad \checkmark \\
 &= \frac{\sinh}{\sinh}
 \end{aligned}$$

Q14

a) (ii) $\frac{d}{dx} [\tan x] = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h}$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{\sinh}{\cos x \cos(x+h) \cdot h} \quad \checkmark \\
 &= \lim_{h \rightarrow 0} \frac{\sinh}{h} \cdot \frac{1}{\cos x \cos(x+h)} \\
 &= 1 \times \frac{1}{\cos x \cos x} = \frac{1}{\cos^2 x} = \sec^2 x
 \end{aligned}$$

as $\lim_{h \rightarrow 0} \frac{\sinh}{h} = 1$

(Need to show all necessary steps to get 1 m)

alt solution to (a)(i)

$$\begin{aligned}
 \tan(x+h) - \tan x &= \frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x} \\
 &= \frac{\sin(x+h)\cos x - \sin x \cos(x+h)}{\cos x \cos(x+h)} \\
 &= \frac{\sin[(x+h) - x]}{\cos x \cos(x+h)} \\
 &= \frac{\sinh}{\cos x \cos(x+h)}
 \end{aligned}$$

Q14)

$$\begin{aligned}
 \text{b)(i)} \quad & (1+x)^{2n} \left(1 - \frac{1}{x}\right)^{2n} \\
 &= \left[\left(1+x\right) \left(1 - \frac{1}{x}\right) \right]^{2n} \\
 &= \left[1 - \frac{1}{x} + x - 1 \right]^{2n} \\
 &= \left(x - \frac{1}{x} \right)^{2n} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) RHS, } T_{r+1} &= \binom{2n}{r} x^{2n-r} \left(-\frac{1}{x}\right)^r \\
 &= \binom{2n}{r} x^{2n-r} (-1)^r (x)^{-r} \\
 &= \binom{2n}{r} x^{2n-2r} (-1)^r \\
 \text{Let } r=n, \text{ then } \binom{2n}{n} (-1)^n &\quad \checkmark \\
 &\text{is the coef. of the constant term.}
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS, } & \left[\binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + \binom{2n}{2n}x^{2n} \right] x \\
 & \left[\binom{2n}{0} - \binom{2n}{1}\left(\frac{1}{x}\right) + \binom{2n}{2}\left(\frac{1}{x^2}\right) - \dots - \binom{2n}{2n}\left(\frac{1}{x}\right)^{2n} \right]
 \end{aligned}$$

$$\therefore \text{Constant term} = \binom{2n}{0}\binom{2n}{0} - \binom{2n}{1}\binom{2n}{1} + \binom{2n}{2}\binom{2n}{2} - \dots + \binom{2n}{2n}\binom{2n}{2n} \quad \checkmark$$

Equating constants from both sides,

$$\binom{2n}{n}(-1)^n = \binom{2n}{0}^2 - \binom{2n}{1}^2 + \binom{2n}{2}^2 - \dots + \binom{2n}{2n}^2$$

Q14)

$$\text{c) } f(x) = 2x^3 - 3x^2 + 0.999$$

$$\text{(i) } f(1) = 2(1) - 3(1) + 0.999 = -0.001$$

$\therefore$ There is a root close to the root. $\checkmark$

$$\text{(ii) } f'(x) = 6x^2 - 6x$$

$$\text{at } x=1, f'(1) = 6(1)^2 - 6(1) = 0 \quad \checkmark$$

It is a stationary point, $\therefore$ there is no $\checkmark$
tangent which will intersect the x-axis.

$$\text{(iii) } f'(x) = 6x^2 - 6x$$

$$f'(1.5) = 4.5$$

$$f(1.5) = 0.999$$

$$x_2 = 1.5 - \frac{f(1.5)}{f'(1.5)} = 1.278 \quad \checkmark \quad (\text{calculator readout})$$

Q14)

$$d) \sum_{j=1}^n \sin(2j-1)x = \frac{1 - \cos 2nx}{2 \sin x}$$

Step 1: Test $n=1$

$$\text{LHS} = \sin(2(1)-1)x = \sin x$$

$$\text{RHS} = \frac{1 - \cos 2x}{2 \sin x} = \frac{1 - (1 - 2 \sin^2 x)}{2 \sin x} = \sin x$$

It is true for $n=1$

Step 2: Assume that it is true for $n=k$.

$$\sum_{j=1}^k \sin(2j-1)x = \frac{1 - \cos 2kx}{2 \sin x}$$

Step 3: Need to prove that it is true for $n=k+1$

$$\text{i.e.} \sum_{j=1}^{k+1} \sin(2j-1)x = \frac{1 - \cos 2(k+1)x}{2 \sin x}$$

$$\text{LHS} = \sum_{j=1}^k \sin(2j-1)x + \sin(2(k+1)-1)x$$

$$= \frac{1 - \cos 2kx}{2 \sin x} + \sin(2k+1)x \quad (\text{from step 2}) \checkmark$$

$$= \frac{1 - \cos 2kx + 2 \sin x \sin(2k+1)x}{2 \sin x}$$

$$= \frac{1 - \cos 2kx + 2 \sin x \sin(2kx+x)}{2 \sin x}$$

$$= \frac{1 - \cos 2kx + 2 \sin x (\sin 2kx \cos x + \cos 2kx \sin x)}{2 \sin x}$$

$$= \frac{1 - \cos 2kx + 2 \sin x \sin 2kx \cos x + 2 \cos 2kx \sin^2 x}{2 \sin x} \quad \checkmark$$

$$= \frac{1 - \cos 2kx + \sin 2kx \sin 2x + \cos 2kx (1 - \cos 2x)}{2 \sin x}$$

$$= \frac{1 - \cancel{\cos 2kx} + \sin 2kx \sin 2x + \cancel{\cos 2kx} - \cos 2kx \cos 2x}{2 \sin x}$$

$$= \frac{1 - (\cos 2kx \cos 2x - \sin 2kx \sin 2x)}{2 \sin x} \quad \checkmark$$

$$= \frac{1 - \cos(2kx+2x)}{2 \sin x}$$

$$= \frac{1 - \cos 2(k+1)x}{2 \sin x} = \text{RHS}$$

$\therefore$ It is true for $n=k+1$.

As it is true for $n=1$, it must be true for $n=2, 3, \dots$ and so on as it is true for $n=k+1$. $\therefore$ By induction, it is true for all $n \geq 1$.